

Breaking Transposition ciphers

brute force

ciphertext only

known plaintext

chosen plaintext

chosen ciphertext

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chosen plaintext

chosen ciphertext

Small key length only

Hard for large permutations

Match letters together

Choose ABCDE ... and match letters

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known plaintext	Match letters together
chosen plaintext	Choose ABCDE ... and match letters
chosen ciphertext	Choose ABCDE ... and match letters

The seeds of modern cryptography

1. **Diffusion**

Mix-up symbols

Transposition Cipher

2. **Confusion**

Replace a symbol with another

Polyalphabetic Cipher

3. **Randomization**

Repeated encryption of the same text are different

OTP